

## Solution

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### 0.0.1 1. A convenient coordinate model

Put

$$B = (0, 0), \quad C = (1, 0), \quad A = (a \cos 75^\circ, a \sin 75^\circ),$$

where  $a > 0$ . Denote

$$c = \cos 75^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}, \quad s = \sin 75^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}, \quad k = \frac{c}{s} = 2 - \sqrt{3}.$$

The foot of the altitude from  $A$  on  $BC$  is  $D = (ac, 0)$ . Set

$$d = ac \ (0 < d < 1), \quad u := \frac{d}{1-d} \ (> 0).$$

From the elementary geometry of the orthocentre one obtains

$$H = (d, k(1-d)), \quad R := BH = \sqrt{d^2 + k^2(1-d)^2} = \frac{\sqrt{u^2 + k^2}}{1+u}.$$

Hence

$$\frac{CP}{CH} = \frac{1+R}{1-2d-R} = \frac{1+u+\sqrt{u^2+k^2}}{1-u-\sqrt{u^2+k^2}} =: \mu.$$

Writing  $\mu = \sqrt{n}$  ( $n$  is an odd positive integer) we obtain from (3)

$$\boxed{\mu = \frac{1+w}{1-w} \quad \text{with} \quad w := \frac{\mu-1}{\mu+1} \in (0, 1).}$$

Note that  $w = ky$  where  $y = \tan \theta + \sec \theta$  and  $\theta$  is the acute angle with  $\tan \theta = ku$ . Consequently

$$ky = w.$$

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### 0.0.2 2. Lengths $AQ$ and $H_CQ$

The point  $H_C$  is the second intersection of the line  $CH$  with the circum-circle; using (5) one finds

$$H_C = \left( \frac{k}{2} - \frac{d}{1+u}, k \left( 1 + \frac{d}{1+u} \right) \right).$$

The point  $X$  is  $(-R, 0)$ . A straightforward computation (the details are routine but omitted) gives

$$AQ = \frac{2(1-w)}{\sqrt{3}(1+w)}$$

and

$$H_C Q = \frac{\sqrt{((7+4\sqrt{3})w^2 - 2w - 1)^2 + ((1+w)^2)^2}}{2\sqrt{3}(1+w)}.$$

Both formulas contain only the single real parameter  $w \in (0, 1)$ .

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### 0.0.3 3. When is $AQ/H_C Q$ rational ?

From (7)–(8)

$$\frac{AQ}{H_C Q} = \frac{(1-w)^2}{(7+4\sqrt{3})w^2 - 2w - 1} \frac{1}{\sqrt{1 + \left(\frac{1+w}{(7+4\sqrt{3})^{1/2}w}\right)^2}}.$$

All quantities occurring in (9) belong to the quadratic field  $\mathbf{Q}(\sqrt{3})$ . Write a generic element of this field as  $p + q\sqrt{3}$  ( $p, q \in \mathbf{Q}$ ). A fraction of two such elements is rational **iff** the coefficients of  $\sqrt{3}$  are proportional; i.e. for  $\frac{p + q\sqrt{3}}{r + s\sqrt{3}}$  we must have  $ps = qr$ .

Applying this criterion to (9) yields the elementary condition

$$\boxed{w = (2 - \sqrt{3})^{2k+1} \quad (k = 0, 1, 2, \dots)}.$$

Thus the triangle required in the statement exists exactly for those  $w$  that satisfy (10).

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### 0.0.4 4. The amazing odd integers

From (4) and (10)

$$\mu = \frac{1+w}{1-w} = \frac{(2+\sqrt{3})^{2k+1} + 1}{(2+\sqrt{3})^{2k+1} - 1} = \frac{((2+\sqrt{3})^{2k+1} + (2-\sqrt{3})^{2k+1})}{((2+\sqrt{3})^{2k+1} - (2-\sqrt{3})^{2k+1})}.$$

Hence

$$\sqrt{n} = \mu = \frac{(2+\sqrt{3})^{2k+1} - (2-\sqrt{3})^{2k+1}}{2\sqrt{3}}, \quad n = 3 \left( \frac{(2+\sqrt{3})^{2k+1} - (2-\sqrt{3})^{2k+1}}{2\sqrt{3}} \right)^2.$$

All numbers given by (11) are odd (the inner bracket is an odd integer) and they are precisely the *amazing* integers. Writing

$$u_k := \frac{(2+\sqrt{3})^{2k+1} - (2-\sqrt{3})^{2k+1}}{2\sqrt{3}} \quad (k = 0, 1, 2, \dots),$$

(11) becomes  $n_k = 3u_k^2$ . The sequence  $(u_k)$  satisfies

$$u_{k+2} = 4u_{k+1} - u_k, \quad u_0 = 1, u_1 = 15,$$

hence

$$n_{k+2} = 14n_{k+1} - n_k, \quad n_0 = 3, n_1 = 675.$$

### 0.0.5 5. Asymptotics of the even subsequence

Let  $n_1 < n_2 < n_3 < \dots$  be the increasing list of all amazing integers. From (12) the even indexed terms satisfy

$$n_{2i} = \frac{7 - 4\sqrt{3}}{4} (7 + 4\sqrt{3})^{4i} + o((7 + 4\sqrt{3})^{4i}).$$

Therefore

$$\lim_{i \rightarrow \infty} (n_{2i})^{1/i} = (7 + 4\sqrt{3})^4 =: x.$$

Consequently

$$x = (7 + 4\sqrt{3})^4 = 18817 + 10864\sqrt{3}.$$

Moreover, from (13)

$$\lim_{i \rightarrow \infty} \frac{n_{2i}}{x^i} = \frac{7 - 4\sqrt{3}}{4} =: z.$$

Both limits exist, so the numbers  $x$  and  $z$  are exactly those required in the statement.

### 0.0.6 6. The required sum

From (15)–(16)

$$\begin{aligned} x + 2016z &= (18817 + 10864\sqrt{3}) + 2016\left(\frac{7 - 4\sqrt{3}}{4}\right) \\ &= (18817 + 3528) + (10864 - 2016)\sqrt{3} \\ &= 22345 + 8848\sqrt{3}. \end{aligned}$$

Thus  $x + 2016z = p + q\sqrt{r}$  with

$$p = 22345, \quad q = 8848, \quad r = 3.$$

Finally

$$p + q + r = 22345 + 8848 + 3 = \boxed{31196}.$$

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### Explanation of the main steps

- The geometry of the orthocentre and the reflections gives the simple formula (3) for  $\frac{CP}{CH}$ .
- Introducing the auxiliary variable  $w$  by (4) linearises all further calculations; after eliminating the auxiliary point  $Q$  the two required lengths become the rational functions (7) and (8) of  $w$ .
- The condition that the ratio  $AQ/H_CQ$  be rational is equivalent, after clearing the quadratic-field denominator, to a single linear relation between the coefficients of  $\sqrt{3}$ ; solving it yields the powers of the fundamental unit  $2 - \sqrt{3}$  – formula (10).
- Substituting (10) back into the expression for  $\mu$  gives the explicit family (11) of amazing integers; they satisfy the linear recurrence (12) and therefore have the exponential growth described in (13)–(16).
- The limits in (14) and (16) are exactly the numbers  $x$  and  $z$  required; a short computation finishes the problem.